



Most battery chargers stop charging the battery when it attains its maximum charging voltage set by the circuit. This circuit charges the battery at a particular voltage, that is, absorption voltage, and once the maximum charging voltage is attained, the charger changes the output voltage to float voltage for maintaining the battery at that voltage. Absorption and floating voltages are dependent on the type of battery.

For this charger, voltages are set for a sealed lead-acid (SLA) 12V, 7Ah battery, for which absorption voltage is 14.1V to 14.3V and floating voltage is 13.6V to 13.8V. For safe working and to avoid overcharging of battery, absorption voltage is selected as 14.1V and floating voltage is selected as 13.6V. These values are to be set as specified by the battery manufacturer.

Circuit and working

Circuit diagram of the 12V battery absorb and float charger is shown in Fig. 1. It is built around step-down transformer X1, adjustable voltage

regulator LM317 (IC1), op-amp comparator LM358 (IC2) and a few other components.

The 230V AC primary to 15V-0-15V, 1A secondary transformer used in this circuit steps down mains voltage, which is rectified by diodes D1 and D2 and smoothened by capacitor C1. This voltage is given to the input of LM317 for regulation.

The basic circuit is a regulated power supply using LM317, with a control on output by changing resistance at adjust pin 1. A good heat-sink is required for LM317. LM358 is a dual-operation amplifier that is used here to control overcharging of the battery. Capacitor C4 should be as near as possible to pin 1 of IC2.

Jumper J1 is used for calibration (set-up). While setting the charging voltage, remove the jumper and connect it back after calibration.

Test Points

Test point	Details
TP0	GND
TP1	Around 0.5V
TP2	13.6V-14.1V
TP3	High when battery is fully charged

Semiconductors:

- | | |
|-----------|--------------------------------------|
| IC1 | - LM317 adjustable voltage regulator |
| IC2 | - LM358 op-amp |
| T1 | - BC547 npn transistor |
| LED1-LED3 | - 5mm LED |
| ZD1 | - 6.8V zener diode |
| D1-D5 | - 1N4007 rectifier diode |

Resistors (all 1/4-watt, $\pm 5\%$ carbon, unless stated otherwise):

- | | |
|--------|------------------------|
| R1 | - 270-ohm |
| R2 | - 2.2-kilo-ohm |
| R3, R6 | - 10-kilo-ohm |
| R4, R5 | - 22-kilo-ohm |
| R7 | - 0.2-ohm, 5W |
| R8, R9 | - 4.7-kilo-ohm |
| VR1 | - 2-kilo-ohm potmeter |
| VR2 | - 5-kilo-ohm potmeter |
| VR3 | - 20-kilo-ohm potmeter |

Capacitors:

- C1 - 2200 μ F, 40V electrolytic
C2, C3 - 10 μ F, 25V electrolytic
C4 - 0.1 μ F ceramic disk

Miscellaneous:

- | | |
|------------|---|
| X1 | - 230V AC primary to 15V-0-15V, 1A secondary transformer |
| CON1, CON2 | - 2-pin connector terminal
- 12V, 7Ah rechargeable battery |
| J1 | - 2-pin connector for jumper J1 |
| S1, S2 | - On/off switch
- Heat sink for LM317 |

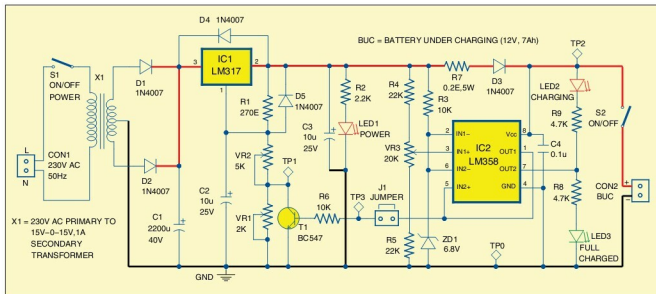


Fig. 1: Circuit diagram of the 12V battery absorb and float charger